ROTATIONAL DYNAMICS

PPT-2

By- Mrs. Prajakta Joshi (S. B. Patil College Ravet)

4) Conical Pendulum:

In pendulum we have a tiny mass attached to a long, flexible, massless, inextensible string, and suspended to a rigid support.

Simple Pendulum:

-- motion of bob is to and fro motion

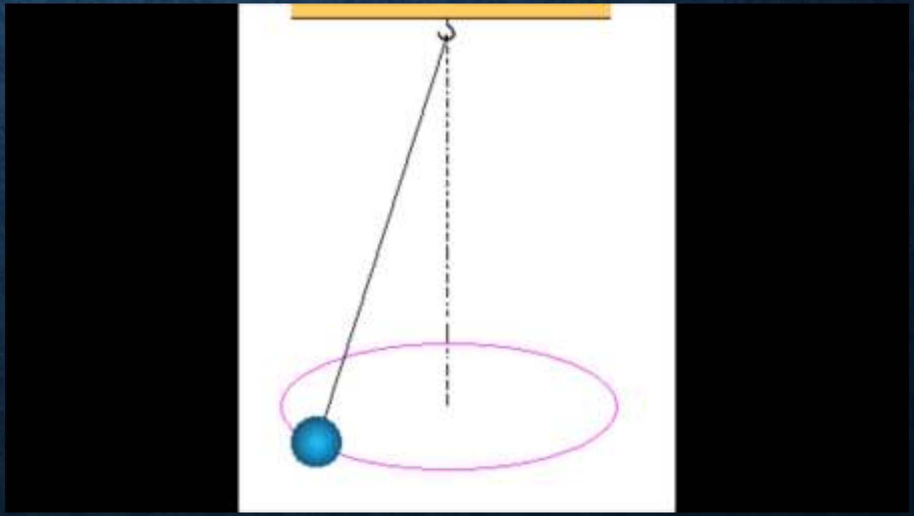
-- oscillates in single vertical plane

Conical Pendulum:

-- motion of bob along a right circular cone of vertical axis

-- bob revolves around the vertical axis

-- bob (point object) performs a horizontal UCM



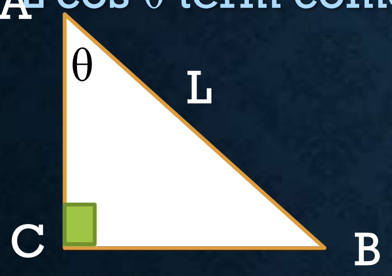
We first study the motion of bob in inertial frame of reference. The bob is assumed to be a point mass.

Mass of bob = m and length of string = L

Forces involved:

1. Weight (mg) vertically downward
2. Tension (T_0) along the string (resolved into components)

How L cos θ term comes?



L cos θ is length and not force.

Consider ABC which is a right angled triangle,

cos θ = adjacent/hypotenuse

cos θ = AC/AB

cos θ = AC/L i.e. AC = L cos θ

Similarly, BC = r = L sin θ

So, L cos θ , *is not any resolved component*

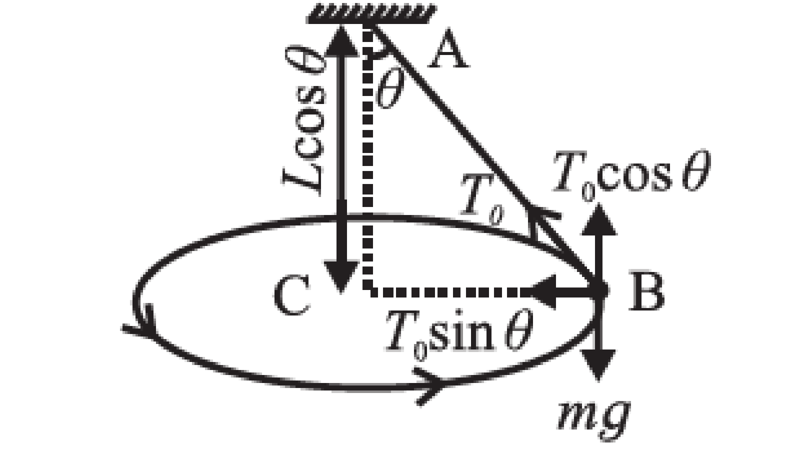


Fig. 1.9 (a): In an inertial frame

From figure,

1. Weight (mg) is balanced by T_0 cos θ at point B i.e. $mg = T_0 \cos\theta$ ----- (1)
2. Centripetal force is given by T_0 sin θ i.e. Centripetal force = $m\omega^2 r = T_0 \sin\theta$ ------- (2)

Because of inertial frame of reference we have not considered centrifugal force here.

We have two equations here $mg = T_0 \cos\theta$ ----- (1)

and resultant force is horizontal and directed towards the centre, $m\omega^2 r = T_0 \sin\theta$ ------- (2)

Dividing (2) by (1) $\dfrac{\omega^2 r}{g} = \dfrac{\sin\theta}{\cos\theta}$ i.e. $\omega^2 = \dfrac{g \sin\theta}{r\cos\theta}$

But $r = L \sin\theta$

Therefore, $\omega^2 = \dfrac{g \cancel{\sin}\,\theta}{L \cancel{\sin}\,\theta \cos\theta}$ we get, $\omega^2 = g / L\cos\theta$

If T is the period of revolution of the bob,

$$\omega = \frac{2\pi}{T} = \sqrt{\frac{g}{L\cos\theta}}$$

$$\therefore \text{Period } T = 2\pi\sqrt{\frac{L\cos\theta}{g}} \qquad \text{--- (1.7)}$$

Frequency of revolution,

$$n = \frac{1}{T} = \frac{1}{2\pi}\sqrt{\frac{g}{L\cos\theta}} \qquad \text{--- (1.8)}$$

The frame of reference of bob is non inertial, therefore, centrifugal force should balance all real forces. This is to bring bob at rest.

So we have , $T_0 \sin\theta = m\omega^2 r$ same in magnitude

as previous equation.

$$\therefore \text{Period } T = 2\pi\sqrt{\frac{L\cos\theta}{g}}$$

Frequency of revolution,

$$n = \frac{1}{T} = \frac{1}{2\pi}\sqrt{\frac{g}{L\cos\theta}}$$

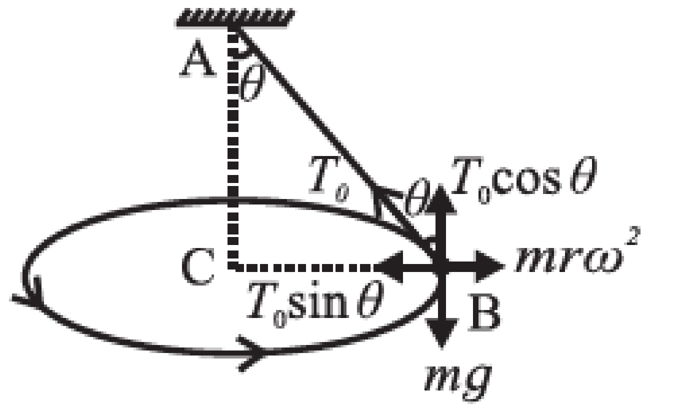


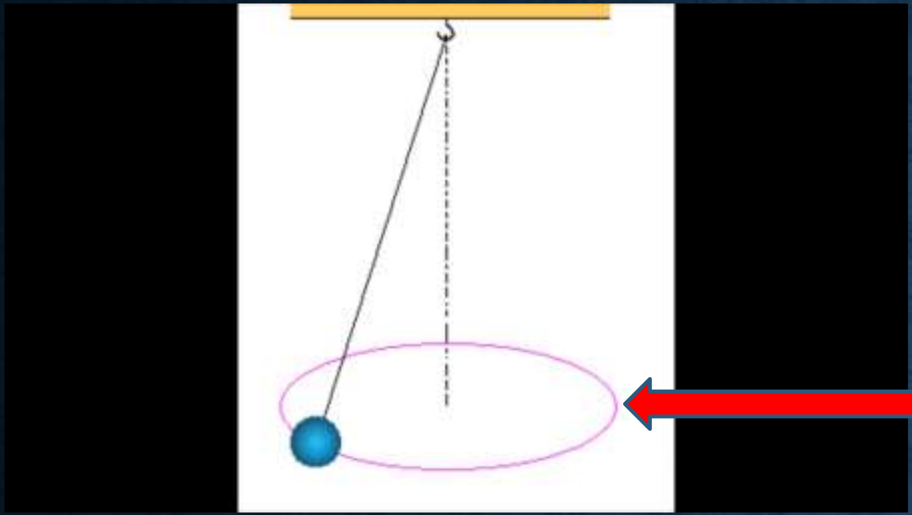
Fig. 1.9 (b): In a non- inertial frame

Points to note:

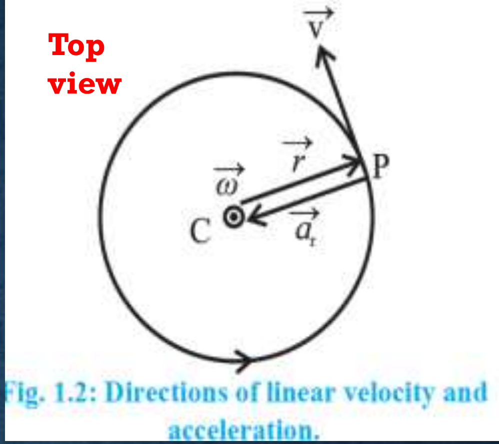
- As L and g are constant, both T and n depends upon θ
- During revolutions, the string can NEVER become horizontal i.e. θ can never be exactly 90°

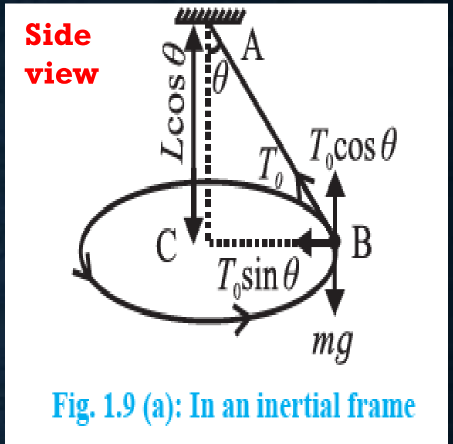
a) If the string is exactly horizontal, no vertical component is there to balance the weight of bob acting vertically downward.

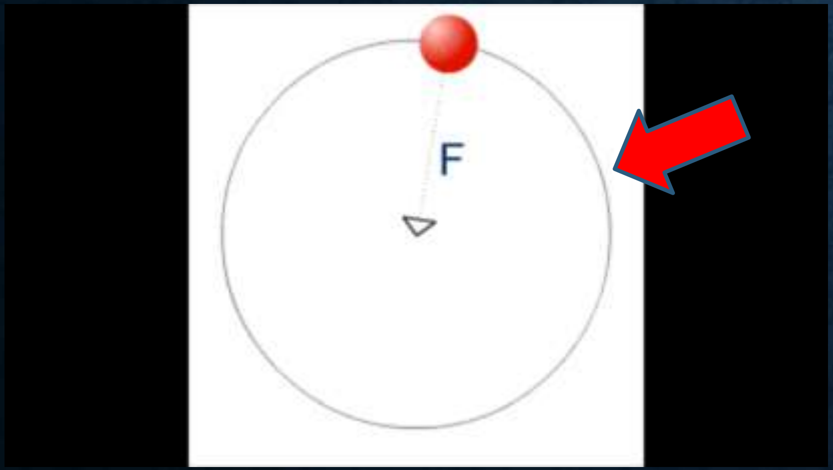
b) Also, if θ = 90°, Time period (T) will be zero and frequency (n) infinite. This will also make Tension in string and thus kinetic energy to be infinite. (NOT POSSIBLE IN REALITY ..!)



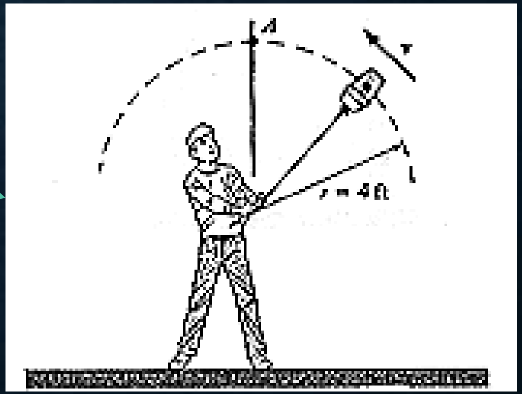
By far, we studied this horizontal circular motion by considering its top-view or side view







In Vertical Circular Motion, we will be studying circular motion like this.
Example: Roller coaster ride
a bucket swung in air



VERTICAL CIRCULAR MOTION

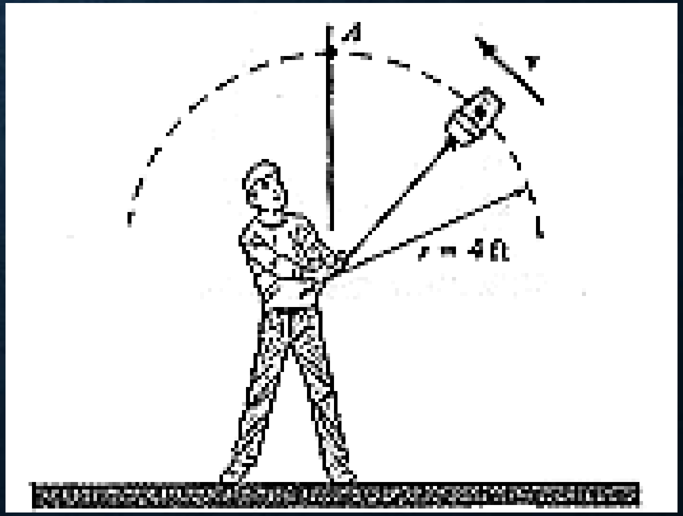
Two types:

1) <u>Controlled Vertical Circular motion</u>:

- Speed is kept constant or is NOT totally controlled by gravity
- Eg. Giant wheel or roller coaster
- We have to supply energy throughout the motion

2) <u>Gravity controlled Vertical Circular Motion</u>

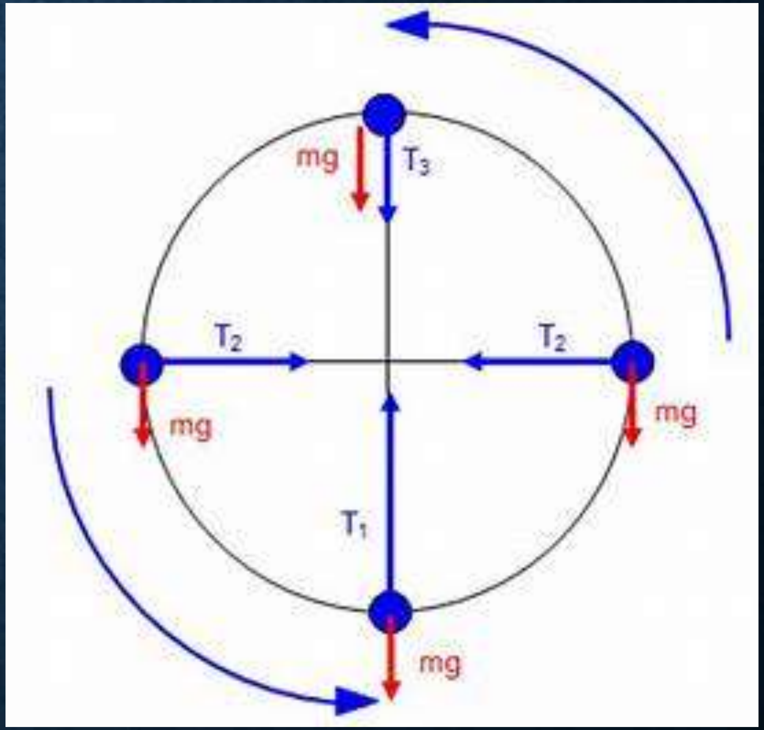
- Motion is controlled only by gravity
- We initially supply energy mostly at lowest point, further entire kinetics (motion without cause) is governed by gravitational force.
- During motion, interconversion of K.E. and gravitational P.E.
- Eg. Mass tied to a string, sphere of death and Vehicle on top of convex over-bridge or swinging bucket



Under the Vertical Circular Motion, we will briefly study dynamics of mass tied to a string and further discuss sphere of death and Vehicle on top of convex over-bridge.

1) **Point Mass undergoing Vertical Circular Motion:** This has two cases, first where mass is tied to a string and second, where mass is tied to a rod.

Case 1: *Mass tied to a string:*



- The bob is considered to be point mass and the string is massless, inextensible and flexible.
- It is whirled along a Vertical Circle, so

bob ----- performs vertical circular motion and

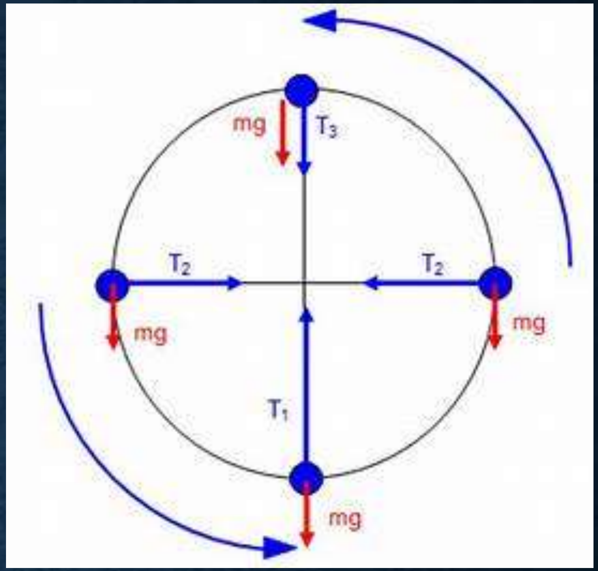
String ---- rotates in vertical plane

- Here only the bob is performing the vertical circular motion so we discuss forces acting on bob only

Forces acting on bob:

1) Weight (mg) vertically downward
2) Force due to tension (T) along the string, always directed towards the centre.

   Magnitude of T changes periodically with time and location



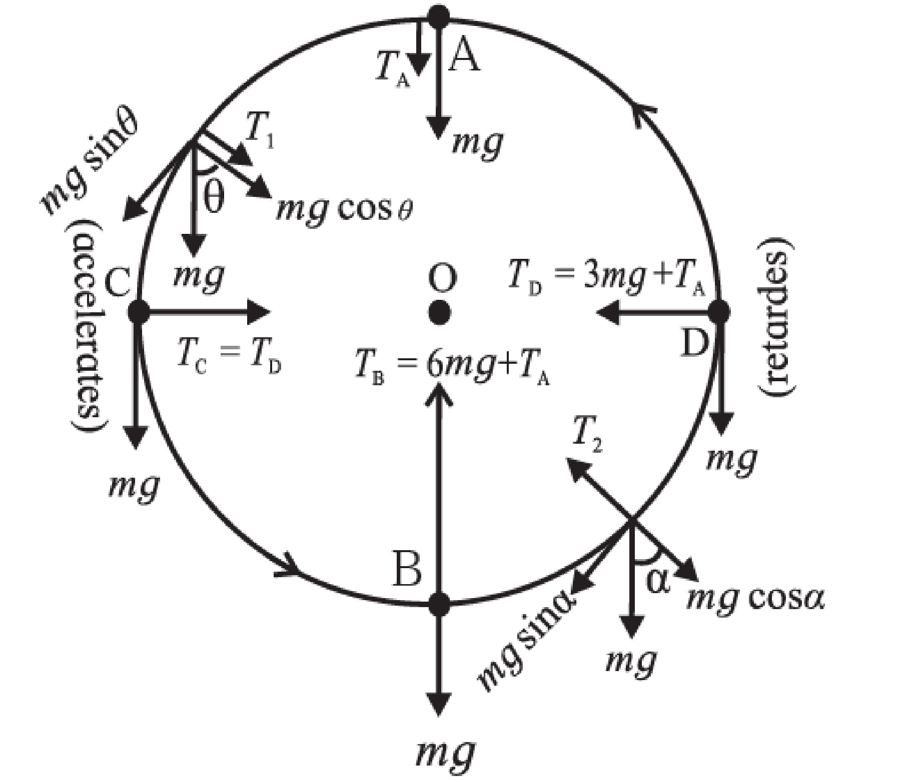


Fig 1.10: Vertical circular motion.

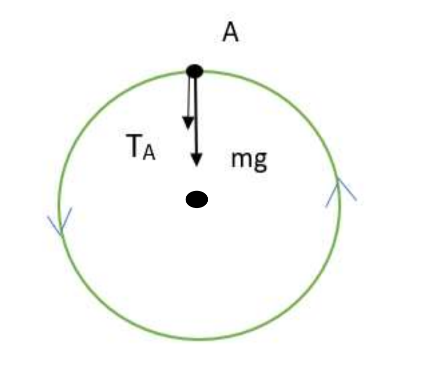
- Because magnitude of T is changing, we consider four different points A, B, C and D on this vertical circle and then analyze each individually.
- Further we have considered two extra points diagonally opposite, whose tension is denoted by T_1 and T_2
- T_1 and T_2 are considered to study tension in string at any random point on vertical circle.

Aim: To find tension T in the string and velocity v at different points under study.

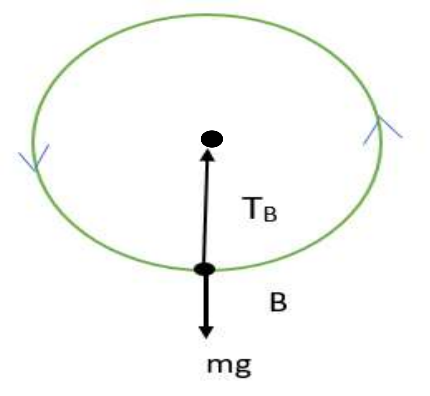
Because the motion is no-uniform, the resultant force due to weight (mg) and tension (T) will not always directed in same direction.

So we study this using four point on the vertical circle

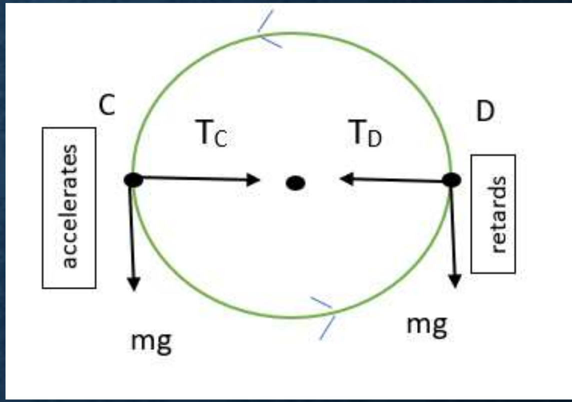
Point A: Uppermost position



Point B: Lowermost



Point C & D: horizontal string



We now write equations for each position:

For A,
mg + T_A = Centripetal force

$$mg + T_A = \frac{mv_A^2}{r}$$

For B,
T_B – mg = Centripetal force

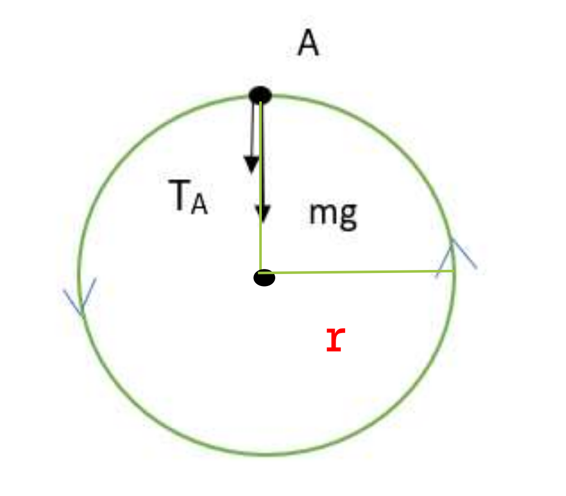
$$T_B - mg = \frac{mv_B^2}{r}$$

For C and D,
$T_C = T_D$ = Centripetal force
(mg only helps in changing direction)

Now, that we have diagram and equations for each we discuss them further individually.

Point A : Uppermost Position

v_A is the speed of particle at A and
r = radius of circle = length of string



- Resultant of mg and T_A provides Centripetal Force

$$mg + T_A = \frac{m v_A^2}{r}$$

Now, for minimum possible speed at this point (i.e. for motion with least energy) , tension T in the string should be zero.

Therefore $T_A = 0$ ------ (1)

i.e. $mg = mv^2/ r$

i.e. $g = v^2/ r$ i.e

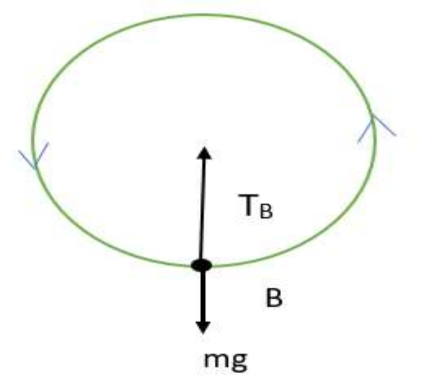
$$\therefore (v_A)_{min} = \sqrt{rg}$$ ------ (2)

Point B : Lowermost Position

- Resultant of mg and T_B provides Centripetal Force.

$$T_B - mg = \frac{mv_B^2}{r}$$

v_B is the speed of particle at B,



- Now, while coming down from A, uppermost point to B, lowermost point displacement = 2r (diameter)
- Also, Motion is governed by gravity only.
- Therefore, the corresponding decrease in P.E. gets converted into K.E.

Decrease in P.E. = Δ K.E.

$$\therefore mg(2r) = \frac{1}{2}mv_B^2 - \frac{1}{2}mv_A^2$$

$$\therefore v_B^2 - v_A^2 = 4rg$$

But we know, $v_A = \sqrt{(rg)}$ therefore, v_B^2 - rg = 4rg

i.e. $(v_B)_{min} = \sqrt{(5rg)}$ ------ (3)

To find equation for tension in string, we consider

$$T_B - mg = \frac{mv_B^2}{r}$$

$$mg + T_A = \frac{mv_A^2}{r}$$

On subtracting (mg + T_A) from (T_B - mg) we get

T_B - mg - mg – T_A = m (v_B^2 - v_A^2) / r

T_B – T_A -2 mg = m (4rg)/r = 4mg

T_B – T_A = 2mg + 4mg

T_B – T_A = 6mg ------ (4)

Derivation for C & D

Going from position A to C,

Change in PE = ΔKE

i.e. $$mgr = \frac{1}{2}mv_C^2 - \frac{1}{2}mv_A^2$$

$$gr = \frac{1}{2}(v_C^2 - v_A^2)$$

$$\boxed{2gr = v_C^2 - v_A^2} \quad \text{—(1)}$$

If we consider $(T_C - T_A)$ i.e.

$$T_C = \frac{mv_C^2}{r} \quad \text{and} \quad T_A = \frac{mv_A^2}{r} - mg$$

$$\therefore T_C - T_A = \frac{mv_C^2}{r} - \frac{mv_A^2}{r} + mg$$

$$= \frac{m}{r}(v_C^2 - v_A)^2 + mg$$

$$= \frac{m}{\cancel{r}}(2g\cancel{r}) + mg$$

$$= 2mg + mg$$

$$\boxed{T_C - T_A = 3mg} \quad \text{—(2)}$$

From (1); $v_C^2 - v_A^2 = 2gr$

But $v_A^2 = rg \Rightarrow v_C^2 - rg = 2gr$

$$\boxed{v_C^2 = 3gr}$$

Arbitrary Positions on Vertical Circle:

- In this, as both weight and Tension are neither along same line or perpendicular. So we resolve them.
- The tangential component of weight (mg sin θ) is used in changing speed.

While going up ------ mg sin θ decreases

While going down ------ mg sin θ increases.

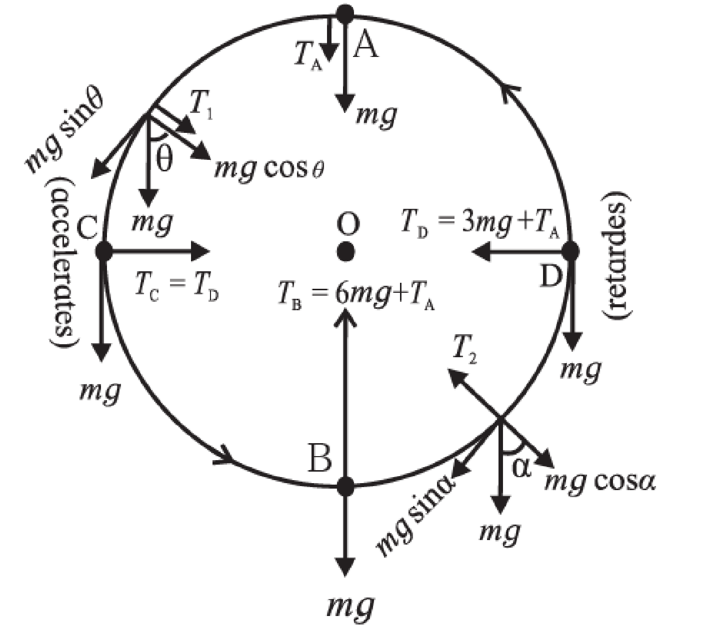


Fig 1.10: Vertical circular motion.

Case 2: Mass tied to a rod:

By far we discussed motion of mass tied to a string. Today we will discuss about the mass tied to a rod.

- So instead of string we have a rod here.
- Now in case of string, there needs to be presence of some Tension acting towards centre, to keep the point mass in motion.
- But since the rod is rigid, we do not need any such Tension to maintain point mass in motion.

For position A: $T_A = 0$ also practically, $v_A = 0$ at uppermost position. This gives us the tension and velocity at A.

For position B, we consider similar method, i.e. Change i … s the difference in their K.E.

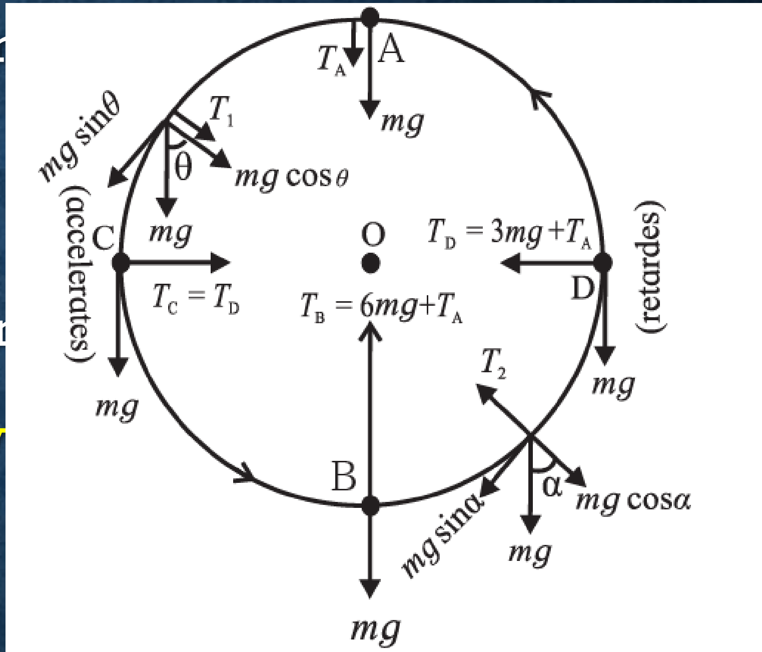


Fig 1.10: Vertical circular motion.

Therefore, we have Change in P.E. = Δ K.E.

i.e. $mg\,(2r) = m\,(v^2_B - v^2_A) / 2$ but $v_A = 0$ therefore, r

Cancelling out m, $2gr = v^2_B/2$ i.e. $v^2_B = 4gr$ i.e. v

For positi … t

$$T_{lowermost} - T_{uppermost} = 6mg$$

$mg\,(r) = m\,(v^2_C - v^2_A) / 2$

But $v_A = 0$ therefore, $mg\,(r) = m\,v^2_C/2$ i.e. $v_C = \sqrt{(2gr)}$

Also, if we take $T_B - T_A =$

2. Sphere of Death in Circus:

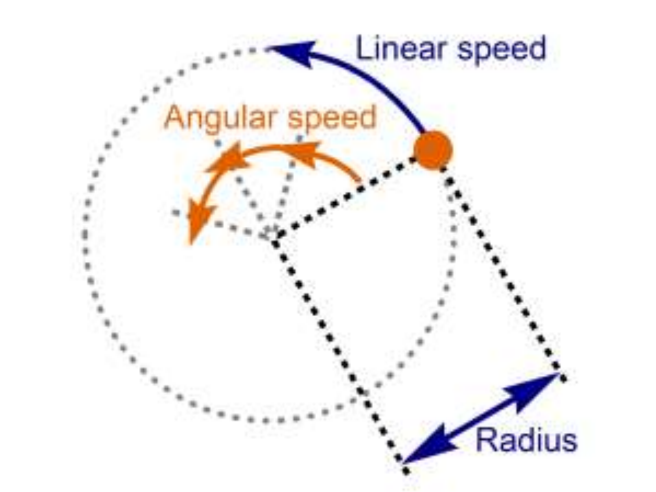
During this, two wheeler riders undergo rounds inside a hollow sphere.

They start with small horizontal circles and eventually perform vertical circles.

Its dynamics is same as a mass tied to a string, just that force due to tension T is replaced by normal reaction N.

Linear speed -------- more for larger circles

Angular speed (frequency) ------ more for smaller circles.



3. Vehicle on top of a convex over-bridge:

You can see the real life example of such a bridge. We intend to study the forces acting on the vehicle when it is at the top.

It only has a part of vertical motion.

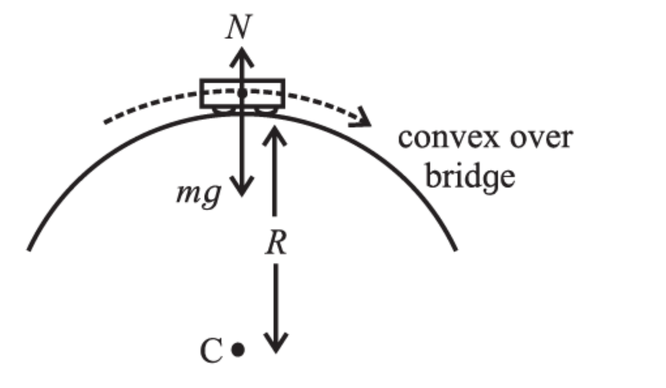


Fig. 1.11: Vehicle on a convex over-bridge.

<u>Forces involved</u>:
1. Weight (mg) acting vertically downward.
2. Normal reaction (N) vertically upward.

- Resultant of these two forces must provide centripetal force.
- Centripetal force acts vertically downwards when vehicle is at uppermost position.
- Therefore we have, $mg - N = mv^2/r$
- Weight should be greater than N to keep the vehicle on road.

Eshima-Ohashi Bridge, Japan.

We have, $mg - N = mv^2/r$

- As v increases, N decreases which may cause the vehicle to go tangentially, that's why we need speed limit here…!
- Normal reaction happens only when two bodies are in contact, thus for safe travel maximum speed limit will be at

  $N = 0$

i.e. $mg = mv^2/r$ i.e. $v^2 = rg$ i.e. $v_{max} = \sqrt{(gr)}$

- This gives the upper limit for speed of vehicle when on bridge
- Another example of similar type is a roller coaster, where we experience the normal reaction N, that keeps us stuck to our seats. It has similar dynamics just like this car.

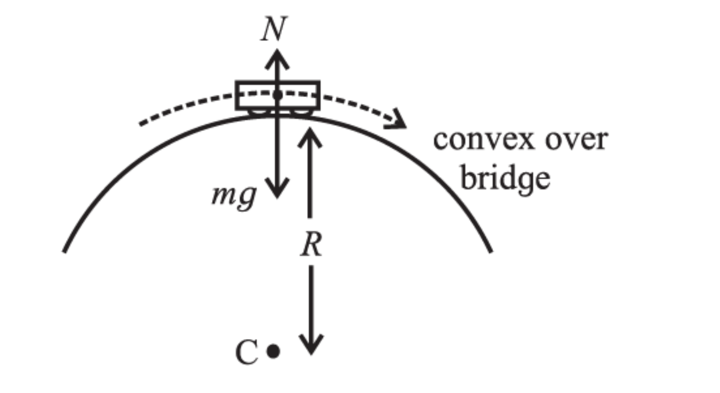


Fig. 1.11: Vehicle on a convex over-bridge.

Moment of Inertia:

It is defined as a body's tendency to resist angular acceleration (thus rotational motion)

There are 3 basic types of M.I.

1. Mass Moment of Inertia → Measure of distribution of mass of object relative to given axis
2. Area Moment of Inertia* → Reflects how body's points are distributed relative to given axis
3. Polar Moment of Inertia* → Shaft or Beam's resistance to being distorted by torsion, as a function of its shape.

Extra information not in syllabus

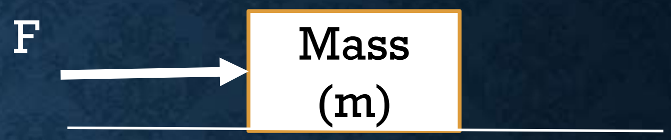
Mass Moment of Inertia:

- It is a measure of distribution of the mass of an object relative to a given axis.
- Denoted by I,
- for single particle $I_O = M R^2$, where O is the axis of rotation & R is the distance from axis
- Unit = $kg\text{-}m^2$ Dimension: $[L^2 M^1]$

MOMENT OF INERTIA

If you recall, When we want to move an object in *lineardirection*, the force applied to move it depends on the mass of object.

(F = ma)



Similarly, in case of rotational motion, eg. Opening a door

The efforts needed to rotate an object depends on

- the mass (m) of object as well as
- distance from axis of rotation (r)

This product of m and r, gives us **Mass Moment of Inertia**

Also called Moment of Inertia (M.I.)

$$I = m \times r^2.$$

- M.I. considers the distance wise distribution of mass around the axis of rotation.
- Like other analogies, M.I. in rotational motion is analogous to mass in linear motion.
- Both of these, express a body's tendency to resist motion (angular or linear)